

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
```

```
#define MAX_USERS 100
#define MAX_NAME 50
#define MAX_EMAIL 50
#define MAX_PASS 20
#define FILENAME "users.txt"
```

```
typedef struct {
    char name[MAX_NAME];
    char email[MAX_EMAIL];
    char password[MAX_PASS];
} User;
```

```
void registerUser() {
    User user;
    FILE *fp = fopen(FILENAME, "a");

    if (fp == NULL) {
```

```
    printf("Error opening file.\n");  
    return;  
}
```

```
printf("Enter name: ");  
scanf(" %[^\\n]", user.name);  
printf("Enter email: ");  
scanf(" %s", user.email);  
printf("Enter password: ");  
scanf(" %s", user.password);
```

```
    fprintf(fp, "%s|%s|%s\\n", user.name,  
user.email, user.password);  
    fclose(fp);  
    printf("User registered successfully.\\n");  
}
```

```
int loginUser() {  
    char email[MAX_EMAIL],  
password[MAX_PASS];  
    User user;
```

```
FILE *fp = fopen(FILENAME, "r");
```

```
if (fp == NULL) {  
    printf("Error opening file.\n");  
    return 0;  
}
```

```
printf("Enter email: ");  
scanf(" %s", email);  
printf("Enter password: ");  
scanf(" %s", password);
```

```
while (fscanf(fp, "%[^|]|%[^|]|%[^\\n]  
\\n", user.name, user.email, user.password) !=  
EOF) {  
    if (strcmp(email, user.email) == 0 &&  
        strcmp(password, user.password) == 0) {  
        printf("Login successful! Welcome,  
%s.\n", user.name);  
        fclose(fp);  
        return 1;
```

```
}  
}
```

```
fclose(fp);  
printf("Login failed. Invalid email or  
password.\n");  
return 0;  
}
```

```
void showAllUsers() {  
    User user;  
    FILE *fp = fopen(FILENAME, "r");  
  
    if (fp == NULL) {  
        printf("No users registered yet.\n");  
        return;  
    }
```

```
    printf("\n--- Registered Users ---\n");  
    while (fscanf(fp, "%[^|]|%[^|]|%[^\\n]  
\\n", user.name, user.email, user.password) !=
```

```
= EOF) {  
    printf("Name: %s, Email: %s\n",  
user.name, user.email);  
}
```

```
fclose(fp);  
}
```

```
int main() {  
    int choice;  
    while (1) {  
        printf("\n1. Register\n2. Login\n3.  
Show All Users\n4. Exit\nChoose an  
option: ");  
        scanf("%d", &choice);  
  
        switch (choice) {  
            case 1: registerUser(); break;  
            case 2: loginUser(); break;  
            case 3: showAllUsers(); break;  
            case 4: exit(0);
```

```
        default: printf("Invalid choice.\n");
```

```
    }
```

```
}
```

```
return 0;
```

```
}
```